The Matplotlib User's Guide

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Chapter 1

antigrain library that ships with matplotlib -

1.5 Interactive

A complete sample rc file is shown in Appendix A.

The matplotlib rc values are read into a dictionary rcParams that contains the key/value pairs. You can changes these values within a script by importing this dictionary. For example, to require that a given script uses numarray, you could do

```
from matplotlib import rcParams
rcParams['numerix'] = 'numarray'
from matplotlib.matlab import
```

Additionally, the commands

numerix settings in setup.py and in your .matplotlibrc file are the same, and are the same as the array package you typically use.

Note that if you install matplotlib anywhere other than the default location, you will need to set the MATPLOTLIBDATA environment variable to point to the install base dir. Eg, if you install matplotlib with python setup.py build --prefix=/home/jdhunter then set MATPLOTLIBDATA to /home/jdhunter/share/matplotlib.

1.7.2 Installing on windows

If you don't already have python installed, you may want to consider using the enthought edition of python, which has scipy, Numeric, and wxpython, plus a lot of other goodies, preinstalled - http://www.enthought.com/python . With the enthought edition of python + matplotlib installer, the following backends should work out of the box: agg, wx, wxagg, tkagg, ps and svg.

Chapter 2

The matlab^{TM} interface

Although matplotlib has a full object oriented API (see Chapter 7), the primary way people create plots is



Figure 2.2: A simple plot decorated with some text labels and an axes grid



Figure 2.3: A sine wave added to the simple plot. This may not be what we wanted. Because the `hold` state was on, the two plots were superimposed.

Using keyword arguments to control line properties

```
title("Don't try this at home, folks", color='#afeeee')
```

b	blue
g	green
r	red
c	cyan
m	magenta
y	yellow
k	black
w	white
0.75	

2.4.2 Loading and saving binary data

ASCII is bloated and slow for working with large arrays, and so binary data should be used if performance is a consideration. To save the array `X` in binary form, use the numerix `tostring` method

```
# open the
```

```
ylabel('Undamped')
```

Likewise, to create two columns and one row of axes, you would use `subplot(121)` to create the left axes and `subplot(122)` to create the right axes. If the total number of axes exceeds single digits, use comma

2.5.3 axes

When you need a finer grained control over axes placement than a□orded by `subplot`, use the funcaxes command. The `axes`

```
# instance methods
>>> l = ylabel(
```



Figure 2.9: Aligning text with *horizontalalignment*, *verticalalignment*, and *multialignment* options to the `text` command; see Listing 2.4

```
ax.text(left, bottom, 'left top',
        horizontalalignment='left',
        verticalalignment='top',
        transform=ax.transAxes)

ax.text(left, bottom, 'left bottom',
        horizontalalignment='left',
        verticalalignment='bottom',
        transform=ax.transAxes)

ax.text(right, top,
```

```
        horizontalalignment='center',
```

2.13.2 toolbar2

The toolbar2 buttons (see Figure 2.15 behave very di□erently from the classic the classic matplotlib toolbar (else why introduce a new one!) despite the visual similarity of the forward and back buttons.

The Forward



Figure 2.15: The newfangled toolbar2, discussed in Section 2.13.2

Then whenever the user clicks anywhere on the figure canvas, your function will be called and passed a `matplotlib.backend`

Chapter 3

Font finding and properties

`matplotlib.fonts.font_manager` is module for finding, managing, and using fonts across-platforms. This module provides a single FontManager that can be shared across backends and platforms. The findfont()

```
family = ['serif', 'sans-serif', 'cursive', 'fantasy',
```

Chapter 4

Collections

Chapter 5

Class	*Summary*
MinuteLocator	locate minutes
HourLocator	locate hours
DayLocator	locate specifed days of the month

stock price, a string formatter (`FormatStrFormatter`) is used to place dollar signs on the y tick labels.

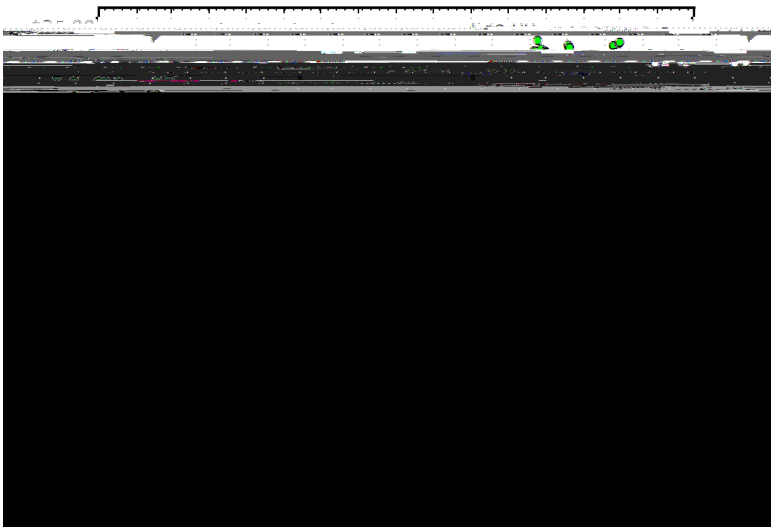


Figure 5.2: Providing custom tick locators and formatters for financial/date plots; see Listing 5.2.

Listing 5.2: Custom date tick locators and formatters; see Figure 5.2

```
import datetime
from matplotlib.matlab import
from matplotlib.dates import MONDAY, SATURDAY
from matplotlib.finance import quotes_historical_yahoo
from matplotlib.dates import MonthLocator, WeekdayLocator, DateFormatter
from matplotlib.tker    import FormatStrFormatter

# the start and end date range fr  the f         plts
```

```
# but we'll override the default with our custom locators and
# formatters
ax.xaxis.set_major_locator(months)
ax.xaxis.set_major_formatter(monthsFmt)
ax.xaxis.set_minor_locator(mondays)

# format the y axis in dollars
ax.yaxis.set_major_formatter(dollarFmt)

# call autoscale to pick intelligent view limits based on our
```

```
y = concatenate( (y1,y2[::-1]) )

p = fill
```

```
im2 = imshow(Z2, cmap=cm.jet, alpha=.9,
             interpolation='bilinear'
```

-

```
gc.set_foreground(
```

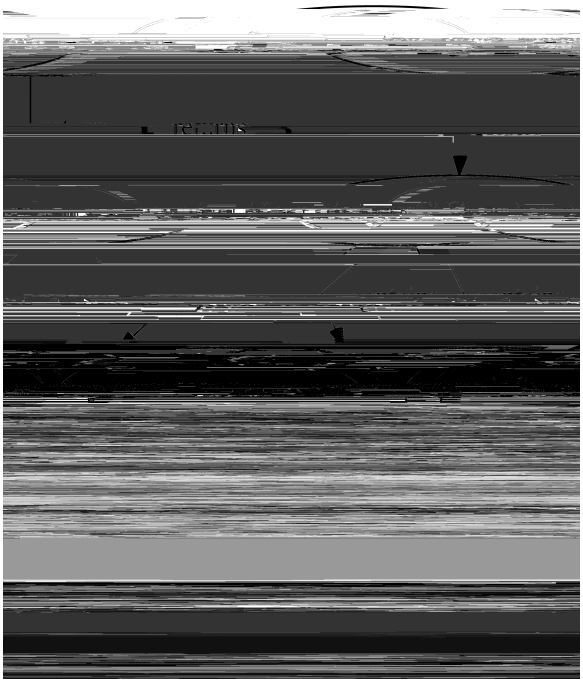


Figure 7.5: The matlab interface function `new_figure_manager` returns a backend dependent concrete implementation of `matplotlib.backend_bases.FigureManagerBase`, which contains the figure canvas and figure

Appendix A

A sample .matplotlibrc

```
#### MATPLOTLIBRC FORMAT
#
# This is a
```

```
lines.markeredgewidth  : 1.0
lines.markersize  : 6         #
```

```
font.stretch          : normal
font.size             : medium
font.serif            : New Century Schoolbook, Century Schoolbook L, Utopia, ITC
```

```
### SAVING FIGURES
# the
```

Appendix C

matplotlib source code license

All of the matplotlib src code is distributed under the Python Software Foundation (PSF) license, which permits commercial and noncommercial free use and redistribution as long as the conditions below are met. The VERSION string below is replaced by the current matplotlib version number with each release.

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Bibliography

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Eric W. Weisstein. *CRC Concise Encyclopedia of Mathematics*. Chapman & Hall/CRC, second edition edition, 2002.